

09/15/2025: Quantifiers

CSCI 246: Discrete Structures

Textbook reference: Sec. 11, Scheinerman

Today's Agenda

- Weekly Quiz ($\approx$ 15 mins)
- Mini-lecture ($\approx$ 15 mins)
- Group exercises ($\approx$ 15 mins)

Subtopics from today's reading

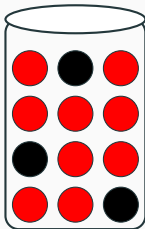
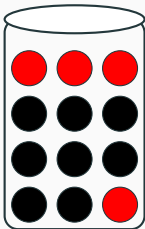
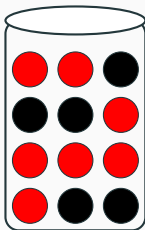
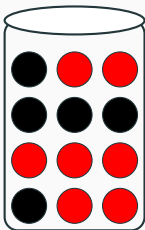
Scheinerman Sec. 11 (Quantifiers) covers the following subtopics

1. Using quantified statements
 - a. How to interpret quantified statements
 - b. How to negate quantified statements
2. Proving quantified statements
 - a. Proving existential statements
 - b. Proving universal statements

Outline for Mini-Lecture:

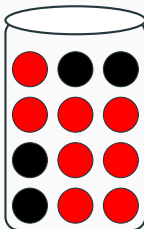
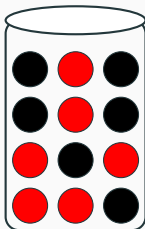
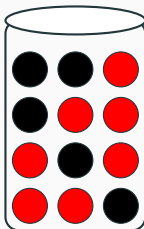
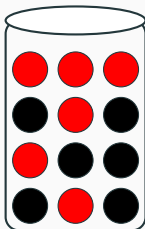
- Using quantified statements
- Proving quantified statements
- Applications

Proposition. $\forall \text{ jar} \in \text{jars}, \exists \text{ ball} \in \text{jar} : \text{ball is red}$



Poll. What is the proposition saying in English?

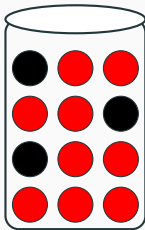
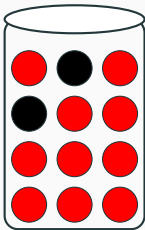
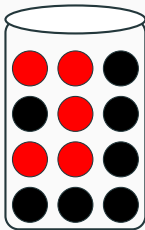
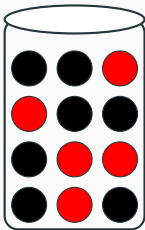
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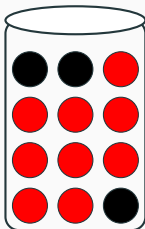
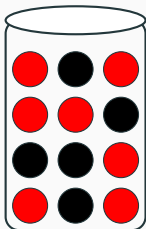
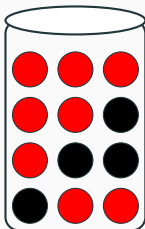
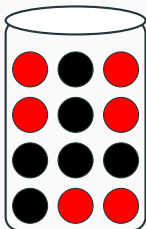
Solution. For each jar in the set of jars, there is a ball in the jar such that the ball is red.

Proposition. $\forall \text{ jar} \in \text{jars}, \exists \text{ ball} \in \text{jar} : \text{ball is red}$



Poll. Is this proposition true or false?

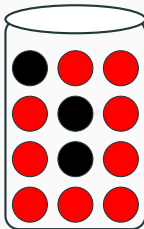
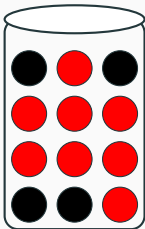
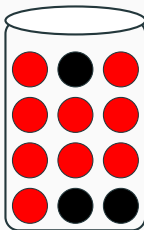
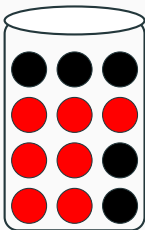
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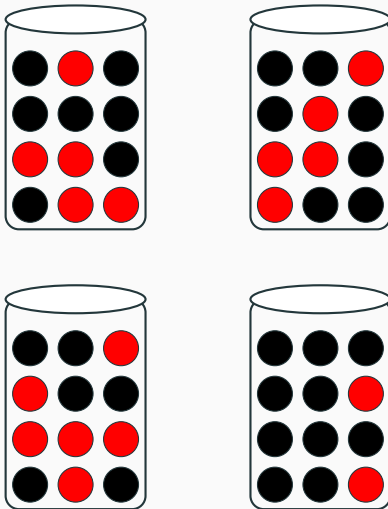
Solution. True

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Poll. How can we change the picture to make the proposition false?

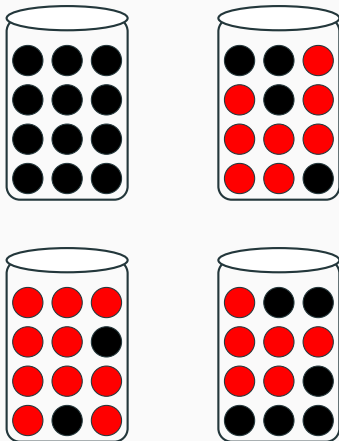
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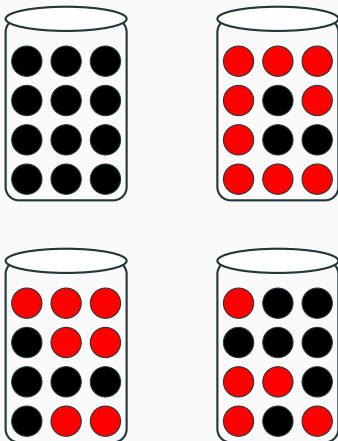
Solution. We need to change one of the jars so that it has no red balls.

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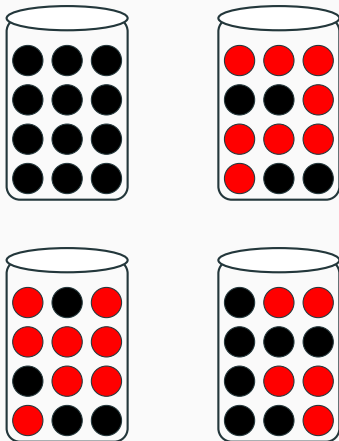
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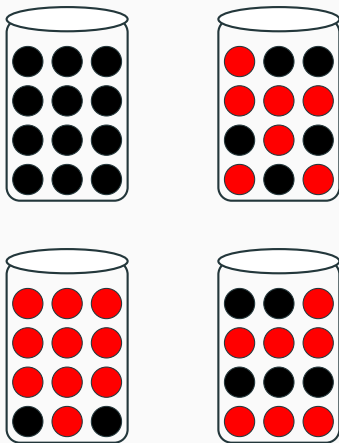
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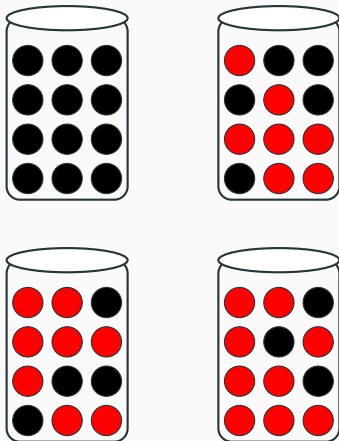
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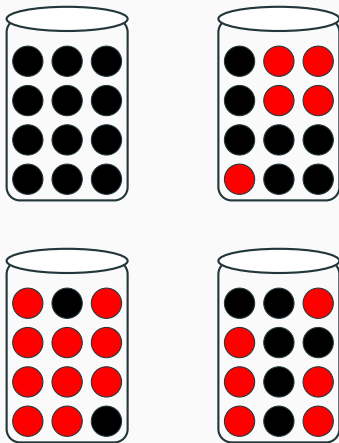
Solution. False.

Proposition. $\forall \text{ jar} \in \text{jars}, \exists \text{ ball} \in \text{jar} : \text{ball is red}$



Poll. How can we change the proposition to make it true?

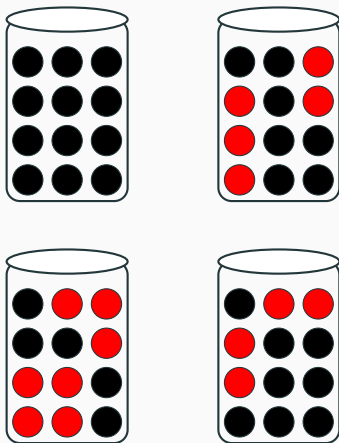
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Poll. How can we change the proposition to make it true?

Solution. $\neg \left(\forall \text{ jar} \in \text{jars}, \exists \text{ ball} \in \text{jar} : \text{ball is red} \right)$

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Poll. How can we change the proposition to make it true?

Solution. $\neg \left(\forall \text{ jar} \in \text{jars}, \exists \text{ ball} \in \text{jar} : \text{ball is red} \right)$

That is, $\left(\exists \text{ jar} \in \text{jars}, : \forall \text{ ball} \in \text{jar} : \text{ball is NOT red} \right)$.

Negating quantified statements

Big Picture. To negate quantified statements, “flip” the quantifiers and carry the “not” operator through.

Negating Existential Statements

$$\neg \left(\exists x \in A, \text{assertions about } x \right) \quad \text{is the same as} \\ \forall x \in A, \neg \left(\text{assertions about } x \right)$$

Negating quantified statements

Big Picture. To negate quantified statements, “flip” the quantifiers and carry the “not” operator through.

Negating Existential Statements

$\neg(\exists x \in A, \text{assertions about } x)$ is the same as
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Negating Universal Statements

$\neg(\forall x \in A, \text{assertions about } x)$ is the same as
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Remark. This technique applies even if there are multiple quantifiers (as we saw in the jars slide).

Negating quantified statements

Big Picture. To negative quantified statements, “flip” the quantifiers and carry the “not” operator through.

Negating Existential Statements

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Negating Universal Statements

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Remark. This technique applies even if there are multiple quantifiers (as we saw in the jars slide). E.g. $\neg \left(\exists \dots : \forall \dots, \exists \dots : \forall \dots, \text{something} \right)$ is the same as $\left(\forall \dots, \exists \dots : \forall \dots, \exists \dots : \text{NOT something} \right)$.

Outline for Mini-Lecture:

- Using quantified statements
- Proving quantified statements
- Applications

Proving existential statements

Proposition. There is an integer that is even and prime.

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Poll. How to express the proposition in quantified form?

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Proposition (with quantifiers). $\exists x \in \mathbb{Z} : x$ is even and prime.

Poll. How to prove the proposition?

Solution. Consider the integer 2. The integer 2 is both even and prime.
(Here you could possibly justify these properties via the definitions.)

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Remark. Proving an existential statement is **akin to finding a counterexample**.

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General strategy.

Proof Template 7 Proving existential statements.

To prove $\exists x \in A$, assertions about x :

Let x be (give an explicit example) ... (Show that x satisfies the assertions.) ... Therefore x satisfies the required assertions. ■

Proving universal statements

Proposition. Every integer that is divisible by 6 is even.

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Proposition (with quantifiers). Let $A = \{x \in \mathbb{Z} : 6 \mid x\}$. Then $\forall x \in A, x$ is even.

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Solution. Let $x \in A$. Then there is an integer y such that $x = 6y$.

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Solution. Let $x \in A$. Then there is an integer y such that $x = 6y$. We can rewrite this as $x = 2(3y)$.

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Poll. How to prove the proposition?

Solution. Let $x \in A$. Then there is an integer y such that $x = 6y$. We can rewrite this as $x = 2(3y)$. Since $3y$ is an integer, x is divisible by 2, and therefore even.

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Remark. Proving an universal statement is **akin to proving an if-then statement**. (“If x is divisible by 6, then x is even”.)

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General strategy.

Proof Template 8 Proving universal statements.

To prove $\forall x \in A$, assertions about x :

Let x be any element of A . . . (Show that x satisfies the assertions using only the fact that $x \in A$ and no further assumptions on x .) . . . Therefore x satisfies the required assertions. ■

Puzzle

Evaluate the (partial) student solution below.

Quiz Question (Quantifiers) from Spring 2025

Let $A = \{x \in \mathbb{Z} : 6|x\}$. Prove that $\forall x \in A$, x is even.

(Partial) Student Solution

$$A = \{x \in \mathbb{Z} : 6|x\} \tag{1}$$

$$\exists y : \forall x, x = 6y \tag{2}$$

Puzzle

Evaluate the (partial) student solution below.

Quiz Question (Quantifiers) from Spring 2025

Let $A = \{x \in \mathbb{Z} : 6|x\}$. Prove that $\forall x \in A, x$ is even.

(Partial) Student Solution

$$A = \{x \in \mathbb{Z} : 6|x\} \quad (1)$$

$$\exists y : \forall x, x = 6y \quad (2)$$

Comments

Line 2 in the solution is False.

$$\underbrace{\exists}_{\text{there exists}} y \underbrace{:}_{\text{such that}} \underbrace{\forall}_{\text{for all}} x, x = 6y$$

(No matter what you pick for y , $6y$ can't represent all integers).

Solution Sketch

General Observation

The primary issue (even to students who scored 4/4) was **communication** – that is structuring the mathematical argument clearly.

Solution Using The If-Then Proof Template.

Proposition. Let $A = \{x \in \mathbb{Z} : 6|x\}$. Then $\forall x \in A$, x is even.

Proof.

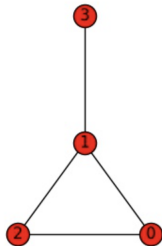
Annotation	Main Text
Convert Prop. to “if-then” form	We show that if $x \in A$, then x is even.
State assumption	Suppose $x \in A$.
Unravel defs.	So $6 x$. That is, there is an integer c such that $x = 6c$.
*** The glue ***	We can write $x = 6c = 2(3c)$.
Unravel defs.	So there is an integer $d(= 3c)$ such that $x = 2d$. In other words, $2 x$.
State conclusion.	So, x is even.

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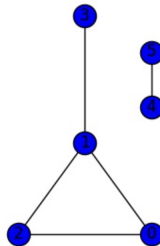
- Using quantified statements
- Proving quantified statements
- **Applications**

Application Ex. 1: Graph Theory

Fully connected graph

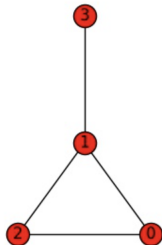


Unconnected graph

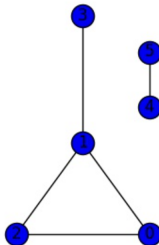


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Unconnected graph



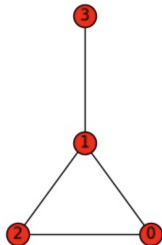
Definition

A graph G is connected if

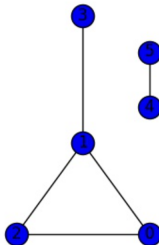
$$\forall u, v \in \text{Vertices}(G), \exists \text{ path } P \text{ from } u \text{ to } v.$$

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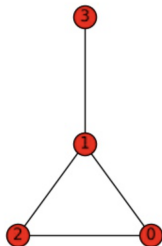
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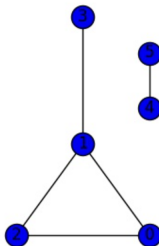
Translation. For all pairs of vertices, there exists a path connecting them.

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Fully connected graph



Unconnected graph



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Translation. For all pairs of vertices, there exists a path connecting them.

Remark. This combines universal and existential quantifiers naturally.

Application Ex. 2: Randomized Algorithms

Imagine you have a randomized algorithm A that tries to solve a problem P , like generating a valid puzzle or finding a satisfiable solution.

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$$\forall x \in \text{ValidInputs}, \forall \epsilon > 0, \exists N \in \mathbb{N} : \\ \Pr[\text{Algorithm } A \text{ solves } x \text{ in } N \text{ steps}] \geq 1 - \epsilon$$

Translation into words. “For all valid inputs x and for any tiny failure probability ϵ , there exists a number of steps N such that the randomized algorithm succeeds with probability at least $1 - \epsilon$.”

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Application. Finding influential samples in large datasets.

(Woj. et al., 2017, *IEEE Big Data*)

